

Controlling the ramification of q along V_F

how the global bound $\text{ram}(\mathcal{P}) \leq d$ together with $G = \mathbb{Z}/p^r$, $r < d$, can supply the input the induction is missing

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June 9, 2026

Purpose. The inductive proof of ?? ($\mathcal{P}^{[\deg n]}$ unramified at η_n for $G = \mathbb{Z}/p^r$) stalls at one point (the red note “*not clear how to finish without analysing the ramification even further*” in `RamificationAfterBlowupIsTame.tex`). The post-mortem in `a_wrong_path.tex` localised the stall to a single, sharp phenomenon: after reducing to

$$\textbf{Goal:} \quad E/F \text{ is unramified at } V_F, \quad I(v_E | V_F) = 1,$$

the hypothesis (H2) only gives $I(v_\star | V_{C'}) = 1$, and the two are linked by an injection

$$I(v_\star | V_{C'}) \hookrightarrow I(v_E | V_F) \tag{1}$$

that need not be onto. The cokernel is *exactly* the ramification of

$$K_{C'}/F \quad \text{at} \quad V_{C'} | V_F,$$

i.e. the ramification that the symmetrisation map $q : (\mathcal{P}_{G/H})^{(d)} \rightarrow (\mathcal{P}_{G/H})^{[d]}$ introduces at the point at infinity. When that ramification is wild it can *absorb* the ramification of E/F , and `a_wrong_path.tex` exhibits a local coincidence $\widehat{K_{C'}} = \widehat{E}$ making (H1)–(H3) hold while the Goal fails. Its closing remark names the cure:

“ruling out such coincidences for a given $(\mathcal{P}, d)$ is exactly the global input the inductive argument lacks.”

This note collects concrete suggestions for supplying that global input by *measuring the ramification of q along V_F* and bounding it by the two global hypotheses we have not yet used:

$$\textbf{(B)} \quad \text{ram}(\mathcal{P} \rightarrow U) \text{ bounded by } d \text{ (i.e. } G^d = 1), \quad \textbf{(C)} \quad G = \mathbb{Z}/p^r, \quad r < d. \tag{2}$$

None of (B), (C) appears in `a_wrong_path.tex`; supplying them is the whole game. [Section 1](#) fixes the object to be measured. [Section 2](#) records three structural observations (in particular: the counterexample lives at $r = d$, and $r < d$ is essentially *automatic* in the totally ramified case). [Section 3](#) gives four attack strategies, ranked. [Section 4](#) proposes the smallest decisive computation.

1 The object to be measured

We keep the notation of `a_wrong_path.tex`: $G = \mathbb{Z}/p^r$ acting totally ramified at p_∞ (after the standard reductions), $H = \ker(G \rightarrow G/H)$ the unique subgroup of order p (the top Artin–Schreier layer), $G/H \cong \mathbb{Z}/p^{r-1}$ the bottom layer, $f : C' \rightarrow C$ the G/H -cover, $p'_\infty \mapsto p_\infty$. Blowing up $d \cdot p_\infty$ and $d \cdot p'_\infty$ gives the DVRs $V_C \subset K(C^{(d)})$ and $V_{C'} \subset K(C'^{(d)})$ at the generic points $\eta_C, \eta_{C'}$ of the exceptional divisors, and

$$K_C = \text{Frac } V_C, \quad F = K((\mathcal{P}_{G/H})^{[d]}), \quad K_{C'} = \text{Frac } V_{C'}, \quad E = K(\mathcal{P}^{[d]G}), \quad V_F = V_{C'}|_F.$$

Definition 1 (the q -defect). Write $w := V_{C'}$, $w_F := V_F = w|_F$. The **ramification of q along V_F** is the ramification of the local extension $\widehat{K_{C'}}/\widehat{F}$ at $w|w_F$: its inertia group $I(w|w_F)$, its ramification index $e(w|w_F)$, its different $\mathfrak{d}_{K_{C'}/F}(w|w_F)$, and (when wild and cyclic) its break β_q in upper numbering. We call any of these data the q -defect.

By (1) and the diagram of `a_wrong_path.tex`, the Goal is equivalent to surjectivity of (1), and the obstruction is precisely a nontrivial q -defect that is “aligned” with E/F . Three reformulations of the Goal, each a possible target:

1. **(Vanishing.)** q is unramified along V_F , i.e. $I(w|w_F) = 1$. Then (1) is an isomorphism and (H2) $\Rightarrow$ Goal immediately. *Strongest; may be false in general — see Section 2(O2).*
2. **(Strict break separation.)** E/F and $K_{C'}/F$ are cyclic wildly ramified of breaks β_E, β_q at V_F with $\beta_q \neq \beta_E$; more usefully $\beta_q < \beta_E$. Then $\widehat{K_{C'}} \neq \widehat{E}$, the local compositum $\widehat{E}\widehat{K_{C'}}$ is genuinely degree p over $\widehat{K_{C'}}$ and ramified, contradicting (H2) unless E/F was unramified to begin with. *This is the cleanest sufficient condition; it directly kills the coincidence $\widehat{K_{C'}} = \widehat{E}$ of `a_wrong_path.tex`.*
3. **(Conductor/different bound.)** $\mathfrak{d}_{K_{C'}/F}(w|w_F)$ is too small to absorb $\mathfrak{d}_{E/F}(v_E|V_F)$ in the tower $F \subset K_{C'} \subset E \cdot K_{C'}$ versus $F \subset E \subset E \cdot K_{C'}$. *Quantitative form of (2); good for an inequality proof.*

The whole point is that all three are statements about q *alone* (the G/H -cover and d), to be bounded by (B)+(C). They do not require re-opening the absorption; they forbid it.

2 Three structural observations

(O1) The counterexample sits exactly at the boundary $r = d$ — (C) excludes it. The fatal coincidence in `a_wrong_path.tex`, ??, is built with $G = \mathbb{Z}/p^2$ (so $r = 2$) and the degree-2 symmetric power ($d = 2$): it is an $r = d$ phenomenon. The tame companion in `inertia_symmetry_obstruction.tex` (eq. “ $d \cdot G_0$ ”) fails the count $|G_0| \mid d$ at $d = |G_0|$, again the boundary. The hypothesis $r < d$ in (2) is precisely the first regime *not* covered by either counterexample. So the right reading of the task is: *prove the obstruction is a boundary ($r \geq d$) phenomenon and disappears for $r < d$* . This is encouraging — (C) is not an arbitrary convenience; it is the complement of the known bad locus.

(O2) Do not aim for “ q unramified” — aim for a break bound. The leading-order (inertia / tame-model) computation of `inertia_symmetry_obstruction.tex` already shows q is genuinely ramified along $V_{C'}$ in general: the diagonal of the E^d -action survives, $I_{V_{C'}}(K_{C'}/F) \cong$

$\mathbb{Z}/\gcd(d, |G/H|)$ in the tame toy model, and $J \supseteq G^d$ in the wild model. So target (1) of [Section 1](#) is too much to hope for. The realistic targets are (2) and (3): bound the *break* of the q -defect strictly below that of E/F . Higher ramification, not inertia, is the right invariant — exactly the moral of `inertia_symmetry_obstruction.tex`.

(O3) $r < d$ is essentially *free* in the totally ramified case. For a totally ramified $\mathbb{Z}/p^r$ extension in which every layer is genuinely wild, the upper-numbering breaks $u_1 \leq \dots \leq u_r$ are integers (Hasse–Arf) and strictly increase, $u_1 < u_2 < \dots < u_r$, whence $u_r \geq u_1 + (r-1) \geq r$. The bound (B) is $G^d = 1$, i.e. $u_r < d$. Therefore

$$r \leq u_r < d,$$

so $r < d$ holds *automatically* once $\mathcal{P}$ is totally ramified with bound d . Two consequences: (i) carrying (C) as a hypothesis through the induction costs nothing in the case that matters; (ii) the genuinely dangerous regime $r \geq d$ can only occur with *coinciding* breaks $u_i = u_{i+1}$, i.e. when some intermediate layer is *not* wildly ramified — which is the degenerate situation the counterexamples exploit. This sharpens the goal: the proof should *use* the strict increase of the u_i .

3 Four strategies

Strategy A (recommended): compute the q -defect in the explicit weighted-blowup local model, in terms of the breaks u_i and d

This is the direct continuation of the method that *already works* for $r = 1$ (??, via ?? and the weighted blowup), and of the local model set up in `inertia_symmetry_obstruction.tex`.

Set-up. Take $k = \bar{k}$, $\widehat{K_C} = k((t))$, uniformiser t at p_∞ . Present $\mathcal{P}$ at p_∞ as an Artin–Schreier–Witt $\mathbb{Z}/p^r$ -extension by a Witt vector $\vec{a} = (a_0, \dots, a_{r-1})$ over $k((t))$, normalised so that the upper breaks are $u_1 < \dots < u_r < d$ (O3). The bottom layer $G/H = \mathbb{Z}/p^{r-1}$ is given by $(a_0, \dots, a_{r-2})$ (breaks $u_1, \dots, u_{r-1}$); the top layer $H = \mathbb{Z}/p$ adds a_{r-1} (break u_r). On the curve C' , a uniformiser u at p'_∞ satisfies $t = u^{p^{r-1}} \cdot (\text{unit})$ (totally ramified G/H -cover).

Symmetric coordinates and the blowup valuation. Use the d copies $t_1, \dots, t_d$ (resp. $u_1, \dots, u_d$) and the elementary symmetric functions $e_1, \dots, e_d$. By the thesis’s own blowup computation (`RamificationAfterBlowupIsTame.tex`, affine case: $\nu_E(f) = \text{ord}_0 f$) and the weighting of `inertia_symmetry_obst`

$$\nu(e_i) = i, \quad \tilde{\nu}(u_j) = 1,$$

and V_C (resp. $V_{C'}$) is read off this weighted blowup. A pole t^{-m} of a defining datum, rewritten in the e_i and divided by the blowup scale, acquires *effective order* $\approx m/d$: this is exactly why $r = 1$ works ($m < d \Rightarrow \text{effective order} < 1 \Rightarrow$ the datum is a unit in the residue field $\Rightarrow$ unramified at η_C).

What to compute. The two completions at $w = V_{C'}$:

- $\widehat{E}/\widehat{F}$: the symmetrised *top* layer a_{r-1} . Extract its break β_E at V_F as a function of u_r and d .
- $\widehat{K_{C'}}/\widehat{F}$ (the q -defect): produced by comparing the scheme symmetric power $(\mathcal{P}_{G/H})^{(d)}$ with the torsor symmetric power $(\mathcal{P}_{G/H})^{[d]}$ at $d \cdot p'_\infty$. Extract β_q as a function of the *bottom* breaks $u_1, \dots, u_{r-1}$ and d .

Target inequality. Show

$$\boxed{\beta_q < \beta_E \quad \text{whenever} \quad u_r < d \text{ and } r < d.} \quad (3)$$

By target (2) of [Section 1](#), (3) forbids $\widehat{K_{C'}} = \widehat{E}$ and makes (H2) imply the Goal. The heuristic in its favour: β_E is driven by the *largest* break u_r , while β_q is driven only by the *bottom* breaks $u_{\leq r-1}$ (the map q sees only $\mathcal{P}_{G/H}$), and by (O3) $u_{r-1} < u_r$ strictly. The role of $r < d$ is to keep the Witt-vector *carries* (which can lift effective orders when the d branches collide) below the blowup scale 1, so that no accumulation of the r bottom layers reaches β_E .

Why this is the recommended route. (i) It is the proven method for $r = 1$, pushed one Witt-degree further. (ii) It computes $I(v_E \mid V_C)$ *directly* and never base-changes along q , so the absorption mechanism cannot even be expressed — it is structurally immune to the `a_wrong_path` obstruction. (iii) It produces a number (β_q) , which is what every other strategy ultimately needs.

Risk. The Witt-vector carry analysis in d symmetric variables is the technical heart and is genuinely involved; the clean statement (3) must be checked, not assumed. [Section 4](#) isolates the smallest instance.

Strategy B: conductor / different accounting in the parallelogram

Stay abstract and use only the transitivity of the different. In the two towers

$$\widehat{F} \subset \widehat{E} \subset \widehat{E\widehat{K_{C'}}} \quad \text{and} \quad \widehat{F} \subset \widehat{K_{C'}} \subset \widehat{E\widehat{K_{C'}}},$$

transitivity gives $\mathfrak{d}_{EK_{C'}/F} = \mathfrak{d}_{EK_{C'}/E} + \mathfrak{d}_{E/F} = \mathfrak{d}_{EK_{C'}/K_{C'}} + \mathfrak{d}_{K_{C'}/F}$ (pulled back appropriately). Hypothesis (H2) is $\mathfrak{d}_{EK_{C'}/K_{C'}} = 0$. Hence

$$\mathfrak{d}_{E/F} \text{ (pulled up)} = \mathfrak{d}_{EK_{C'}/E} - \mathfrak{d}_{K_{C'}/F} \text{ (pulled up)}. \quad (4)$$

The Goal is $\mathfrak{d}_{E/F} = 0$. So one must show the right-hand side of (4) vanishes, i.e. that base-changing $K_{C'}/F$ up to E does not *increase* its different: $\mathfrak{d}_{EK_{C'}/E} = \mathfrak{d}_{K_{C'}/F} \cdot (\text{pullback})$. This is automatic when E/F and $K_{C'}/F$ are arithmetically disjoint, e.g. when their breaks differ — recovering (3) as the clean sufficient condition, now phrased via the different. Combine with the Hasse–Arf/Herbrand convexity already sketched in `inertia_symmetry_obstruction.tex`§(higher):

$$\varphi_G(m_r) = \varphi_{G/H}(m_{r-s}) + \frac{\varphi_H(m_r) - m_{r-s}}{p^{r-s}} \quad (\text{convex combination}) \Rightarrow \varphi_G(m_r) < d,$$

to bound the *total* conductor of $\mathcal{P}$ by d and split it across the layers. *Pro*: no symmetric-variable bookkeeping; rests on standard ramification calculus. *Con*: the disjointness/non-increase step is exactly where the absorption hides — you still need an input equivalent to (3), so B is best as the *packaging* of A's output, not a replacement for it.

Strategy C: rule out $\widehat{K_{C'}} = \widehat{E}$ by a global break invariant

Attack target (2) head-on without computing β_q exactly: find a *global* invariant of $K_{C'}/F$ at V_F that is forced by (B)+(C) to differ from the corresponding invariant of E/F . Candidates:

- **Source of the break.** β_E is a break of the H -layer (top), β_q a break of the symmetrised G/H -tower (bottom). If one shows $\beta_q \in \{\text{values produced by } u_1, \dots, u_{r-1}\}$ and $\beta_E \in \{\text{values produced by } u_r\}$, then (O3)'s strict increase $u_{r-1} < u_r$ separates them.

- **Divisibility / residue.** Even if breaks coincide numerically, the leading coefficients (the “ u^{-m} change of variable” of ??) may be forced distinct by the explicit form of q . A coincidence $\widehat{K}_{C'} = \widehat{E}$ requires equality of Artin–Schreier classes in $\widehat{F}/\wp(\widehat{F})$; show $(B)+(C)$ make the class of $K_{C'}/F$ land in a different coset than that of E/F .

Pro: potentially a short, conceptual proof. *Con:* “find the right invariant” is open-ended; likely needs A’s local model to even identify the invariant.

Strategy D (fallback): change the resolution to make $\eta_{C'}$ codimension one for q

The reason the existing tool fails is purely that ?? controls q only in *codimension one* on $C'^{(d)}$, whereas $V_{C'}$ sits over the codimension- d point $d \cdot p'_\infty$. Idea: choose a modified blowup / weighted resolution $X'_{C'} \rightarrow C'^{(d)}$ in which the chosen valuation $\eta_{C'}$ is a codimension-one point already on the source of q (or factor q through a sequence of codimension-one steps), then apply the codimension-one lemma iteratively. This trades a hard local computation for a careful birational geometry of the symmetric power along the small diagonal. *Pro:* reuses a lemma you already have. *Con:* you must check the modified resolution still computes the *same* $V_C \hookrightarrow V_{C'}$ and still detects the modulus bound d ; the weighted blowup was chosen for a reason (the unweighted one “does not even extend V_C to $V_{C'}$ ”, per `inertia_symmetry_obstruction.tex`).

4 Recommended next step: the decisive small computation

Everything hinges on one number, β_q . Compute it in the first open case and check (3):

Test case. $k = \bar{\mathbb{F}}_p$, $G = \mathbb{Z}/p^2$ ($r = 2$), $H = \mathbb{Z}/p$ bottom, $G/H = \mathbb{Z}/p$. Take $\mathcal{P}$ totally ramified at p_∞ with the bottom-layer break u_1 and top-layer break u_2 , $u_1 < u_2$. Choose d with $u_2 < d$ and $r = 2 < d$ — the smallest honest choice is $d = 3$ (so the `a_wrong_path` counterexample’s $d = 2 = r$ is just excluded). Then:

1. Write the $G/H = \mathbb{Z}/p$ cover $C' \rightarrow C$ locally as $y^p - y = t^{-u_1}$; u = uniformiser at p'_∞ , $t = u^p \cdot \text{unit}$.
2. Form $C'^{(3)}$, the elementary symmetric functions e_1, e_2, e_3 of the three branches, and the weighted blowup $\nu(e_i) = i$. Identify $V_{C'}$, $V_F = V_{C'}|_F$, and the local extension $\widehat{K}_{C'}/\widehat{F}$ induced by q . **Read off** β_q .
3. Symmetrise the top layer a_1 (break u_2) the same way; read off β_E .
4. Check $\beta_q < \beta_E$. If it holds, (H2) $\Rightarrow$ Goal for $(r, d) = (2, 3)$; if it fails, the failure pattern tells you whether the true hypothesis is $r < d$, or $p^2 \mid d$, or something finer.

A single clean computation here either (a) proves the first case and gives the template for the induction, or (b) produces a new, $r < d$ counterexample that pins down the genuinely required hypothesis — both are decisive outcomes. It is the cheapest experiment with the highest information content, and it directly resolves the red TODO at `RamificationAfterBlowupIsTame.tex:671`.

Summary of the logic.

1. Goal $\equiv$ surjectivity of (1) $\equiv$ no q -defect aligned with E/F .
2. The missing input is a bound on the q -defect by $(\mathcal{P}, d)$; (B) and (C) are the unused hypotheses that can give it.
3. The known obstructions are $r = d$ boundary phenomena (O1), and $r < d$ is free in the case that matters (O3); so the obstruction should vanish.
4. The decisive quantity is the break β_q ; compute it (Strategy A, Section 4), package the conclusion via the different (Strategy B), and the absorption is forbidden.